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Mathematical model of dust suppression efficiency of ultrasonic dry fog nozzle in coal mine

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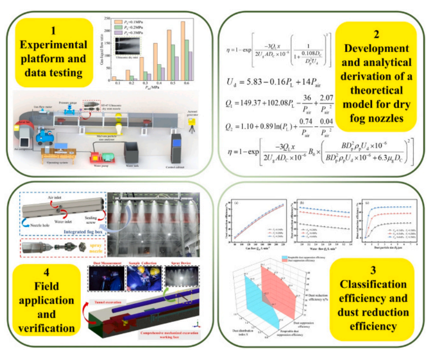
HIGHLIGHTS

- Mathematical model for air-water supply in ultrasonic dry fog nozzles via linear regression.
- Establish predictive model for dust removal efficiency of ultrasonic dry fog nozzles.
- Field test of dry fog nozzle and field verification of dust removal efficiency prediction model.

GRAPHICAL ABSTRACT



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ABSTRACT

To investigate the atomization properties and dust control capability of ultrasonic dry fog nozzles in coal mining environments, an experimental platform was designed to measure key spray parameters. The ultrasonic atomizing nozzle was studied using electromagnetic flowmeter and Malvern spray particle size analyzer. On this basis, a mathematical prediction model for the dust capture efficiency of ultrasonic dry fog nozzles was established. The results show that the air flow rate of the ultrasonic atomization nozzle increases with the increase of the inlet air pressure, while the water flow rate decreases with the increase of the inlet air pressure. In addition, the atomization angle and nozzle Sauter Mean Diameter of the ultrasonic dry fog nozzle decrease with the increase of inlet air pressure, and the nozzle range shows a trend of increasing with the increase of inlet air pressure. The mathematical model of the inlet air flow rate Q_{air}, inlet water flow rate Q_L and dust capture efficiency η of ultrasonic dry fog nozzle was established by linear regression method. Combining the dust particle size and frequency distribution, the theoretical calculation formulas for the overall dust settling efficiency η' and the respirable dust settling efficiency η'' were established, and the effect of the median dust particle size on the dust settling efficiency was analyzed.

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Nomenclature

Q_1	gas flow rate in the experiment
Q_2	liquid flow rate in the experiment
P_{air}	inlet air pressure
P_L	inlet water pressure
D_c	average droplet diameter
V_r	gas-to-liquid slip velocity within the mixing chamber
μ	viscosity coefficient of the liquid
ρ_L	density of the liquid
Q_L/Q_{air}	liquid-gas flow ratio
Q_L	volume flow rates of the liquid
Q_{air}	volume flow rates of the air
σ	liquid surface tension coefficient
V_{air}	gas flow rates
V_L	liquid flow rates
α	angle between the gas and liquid flow rates
V	cross-sectional area
A_L	orifice flow area
A_{air}	area of the air injection hole
U_d	droplet velocity
U_g	airflow velocity
U_{dg}	relative velocity between fog droplets and airflow
B_0	an interception-diffusion experimental constant
B	Cunningham sliding correction coefficient
D_p	diameter of dust particles
ρ_p	density of dust particles
U_0	relative velocity of dust particles and droplets in undisturbed upstream gas
μ_g	gas viscosity coefficient
$R(D_P)$	the sieve retention rate
D_{50}	the mass median diameter of the particulate matter
s	uniformity index for particle size distribution
$f(D_P)$	frequency of dust particles of different sizes
η'	overall dust capture efficiency
η''	respirable dust removal efficiency

1. Introduction

The development of mechanization in the world is gradually accelerating, the mechanization degree of underground coal mining has been continuously improved [1]. Dust generation at coal mine working faces has increased dramatically [2,3]. According to data released by China Health Commission, 11108 new cases of occupational diseases detected in China in 2022. Among them, there were 7577 cases of occupational pneumoconiosis, accounting for 68.21%. Pneumoconiosis patients were mainly distributed in the coal mining industries [4,5]. Dust remains a significant occupational hazard in coal mining [6]. There are many measures to reduce the dust mass concentration in the workplace [7–9]. Spray dust reduction is widely used in coal mines due to its simple operation [10,11] and easy installation. Existing spray systems generally use ordinary nozzles [12,13]. The droplet size of the spray field produced by ordinary nozzles is large [14], the dust reduction efficiency is low [15] and workers are often wet while working [16].

In recent years, ultrasonic dry fog nozzles have been widely used in agricultural and industrial scenarios due to their advantages of small fog particle size, uniform droplet distribution, and good spray effect [17,18]. Ultrasonic atomization is an atomization method that uses ultrasonic waves to break liquid into fine droplets [19]. Compared to pressure-based ordinary nozzles, ultrasonic atomization nozzles require lower water pressure, reduce consumption, and enhance respirable dust control [20,21]. Ultrasonic atomization nozzles are divided into piezoelectric type and fluid type according to the principle of generating ultrasonic waves. The piezoelectric ultrasonic nozzle converts the signal into ultrasonic mechanical vibration using a transducer, and then atomizes the liquid [22–24]. Because of the presence of methane gas in underground coal mines, piezoelectric ultrasonic dry fog nozzles can easily cause safety accidents such as methane explosions. Therefore, fluid-type ultrasonic dry fog nozzles are widely used in underground coal mines. The fluid-type ultrasonic nozzle uses its special structure to convert the kinetic energy of the high-speed fluid at the nozzle outlet into mechanical energy with wave-like vibration, thereby generating ultrasonic waves, which causes the liquid to vibrate and be atomized under the action of ultrasonic waves [25,26]. Ultrasonic atomizing nozzles were first used for fuel atomization. Guo Haoyang et al. [27] investigated the numerical analysis of heat and mass transfer enhancement via ultrasonic atomization technology in the ammonia distillation process. Zhang Yu et al. [28] investigated ultrasonic atomization dynamics: capillary waves, cavitation, and droplet dispersion. Takahisa Kudo et al. [29] studied the phenomenon of atomizing liquids by generating ultrasonic waves in atomizers through numerical simulation, studied the unsteady flow of the nozzle resonance cavity, and classified and summarized the resonance mechanism of the resonance cavity. They classified and summarized the resonance mechanism of the resonant cavity, further improving the resonance mechanism of the fluid dynamic ultrasonic nozzle and promoting the practical application of ultrasonic atomization in industrial production and life [30]. At present, ultrasonic atomizing nozzles have been used for spray dust reduction in coal mines, but there is a lack of systematic theoretical research on them.

The dust capture efficiency of ultrasonic dry fog nozzles is closely related to the nozzle atomization characteristics. Therefore, the authors used a self-designed spray dust suppression test platform to study the flow characteristics, spatial distribution law of atomized particle size and influencing factors of ultrasonic dry fog nozzles. On this basis, a quantifiable predictive mathematical prediction model for the dust capture efficiency of ultrasonic dry fog nozzles was established and the introduction of a dust particle size distribution function for graded calculation of dust efficiency. In order to provide theoretical guidance and pressure setting basis for the application of ultrasonic atomization nozzles in dust prevention and control of underground coal mining working surfaces.

2. Mathematical model of nozzle atomization parameters

2.1. Establishment of mathematical model of nozzle gas-liquid flow

The spray process of ultrasonic dry fog nozzle is a complex two-phase flow process. The calculation of nozzle gas-liquid flow rate involves parameters such as mixing chamber pressure, outlet pressure loss, cross-sectional gas content and gas-liquid two-phase density, which are difficult to calculate using theoretical formulas [31,32]. When the nozzle is working, water enters the mixing cavity from the water injection hole. During this process, the pressure along the process and the mixing chamber pressure are difficult to measure, and the attenuation law is relatively complex. In order to obtain the mathematical model of nozzle gas-liquid flow rate, this study selects the experimental measurement of inlet water pressure of 0.1–0.3 MPa and inlet air pressure of 0.1–0.6 MPa, and uses 1stOpt software to perform regression analysis on the experimental data, and obtains the relationship between the gas-liquid two-phase flow rate of the ultrasonic dry fog nozzle and the inlet water pressure and inlet air pressure.

After preliminary experimental comparison and research by the research team, the ST-47 ultrasonic dry fog nozzle with superior performance was selected [33,34]. The nozzle sprays evenly in a stable state, and the fog flow is conical near the nozzle outlet. The nozzle air

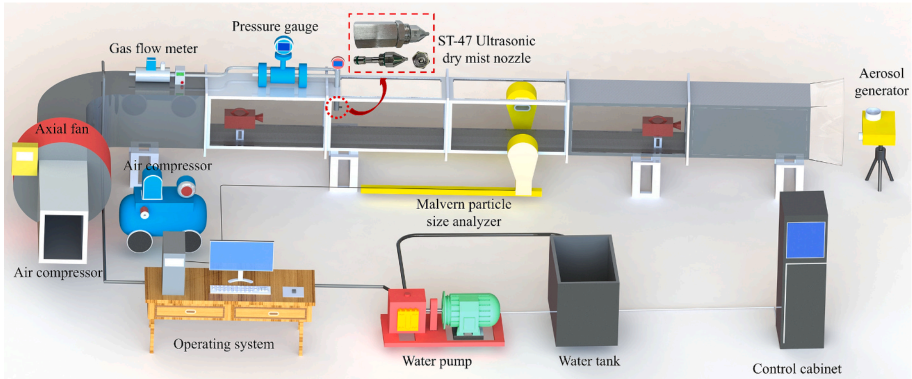

Fig. 1. Ultrasonic dry fog nozzle test system diagram.

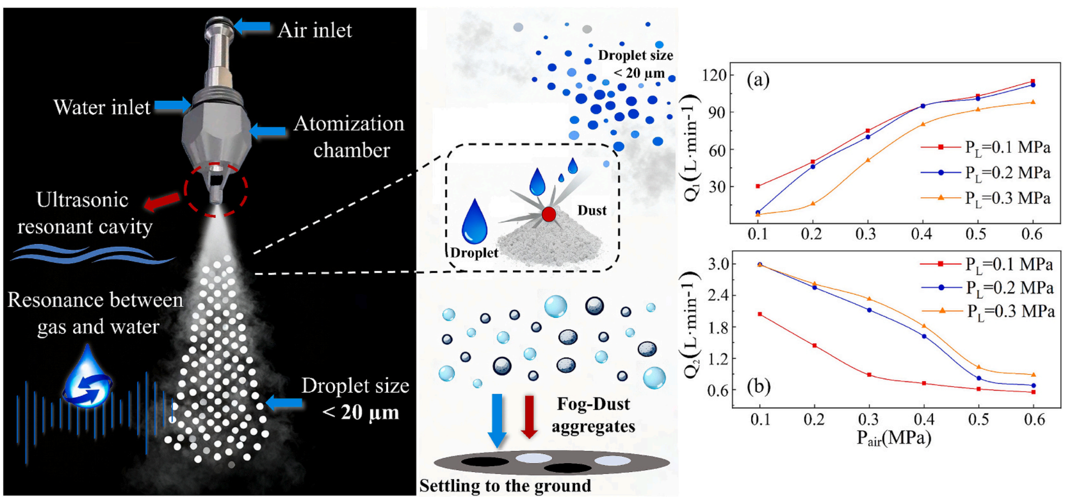

Fig. 2. Nozzle gas-liquid flow diagram (a) Nozzle gas flow (b) Nozzle water flow.

inlet is located at the bottom of the nozzle, with a thread diameter of 12.0 mm. The nozzle water inlet is located on the side of the nozzle, with a thread diameter of 10.0 mm. The spray outlet is located at the front end of the nozzle, and the resonance cavity (ultrasonic generator) is fixed to the front end of the nozzle outlet by a fine steel wire bracket. The nozzle atomization characteristic test system is mainly divided into water and gas supply system. The air supply system is mainly composed of test instruments such as an air compressor air flow meter, an air pressure regulating valve and an air pipe. The water supply system mainly involves instruments such as a BPZ75/12 spray high-pressure water pump (pressure≦12 MPa, rated flow: 4.5 m^3/h), a water tank, a control cabinet, a DX-801XB00150 digital pressure gauge (accuracy is +1.0%) and electromagnetic flowmeter. The ultrasonic dry fog nozzle test system is shown in Fig. 1.

The atomization characteristic parameters of the ultrasonic atomizing nozzle need to be obtained through actual test measurements, and the test condition parameters are determined according to the actual application environment of the nozzle. By measuring the test results of the nozzle atomization characteristic parameters, a mathematical model of the nozzle air supply flow rate, water supply flow rate and droplet particle size is established. Studying the atomization characteristics of the ultrasonic dry fog nozzle can analyze its dust reduction effect on coal dust to a certain extent. Therefore, by measuring the gas-liquid flow rate, atomization angle, range, droplet particle size and other characteristics of the ultrasonic dry fog nozzle, data support can be provided for the establishment of a mathematical model. In the test results, Q_1represents the gas flow rate, Q_2 represents the liquid flow rate, P_{air} represents the inlet air pressure, and P_L represents the inlet water pressure.

Under different inlet water pressures, the gas flow rate of the ultrasonic atomizing nozzle shows a law of gradually increasing with the increase of inlet air pressure. For the fluid type ultrasonic atomizing nozzle, air and water enter the nozzle through the bottom air inlet and the side water inlet along the internal channel, and collide and mix in the nozzle before being ejected from the outlet. As shown in Fig. 2 (a), under the influence of high-frequency vibration, the surface of the liquid film is no longer calm, but rather ripples are generated. In areas where ultrasonic energy is highly concentrated, countless tiny vacuum bubbles are generated in the liquid and collapse instantly. The collapse of these bubbles generates strong local shock waves and microjets, which further help to pulverize the liquid. The internal structure of the nozzle is different, and the energy loss of the gas flow inside is inconsistent, resulting in different nozzle exhaust capacities, which in turn makes the nozzle inlet air pressure the same but the gas flow inconsistent. When the inlet air pressure is 0.6 MPa, the gas flow rate of the ultrasonic atomizing nozzle reaches the maximum value.

The fluid-type ultrasonic atomizing nozzle is a gas-liquid two-phase flow nozzle. The experiment on the flow rate of water from the nozzle was conducted as per the experimental scheme. Fig. 2 (b) presents the experimental results. By comparing the relation between the nozzle's water flow rate and the inlet air pressure, one can find that as the inlet air pressure continues to rise, the back pressure in the mixing chamber increases accordingly, creating significant resistance to the liquid entry. This causes the water flow rate to decrease instead of increase, and when

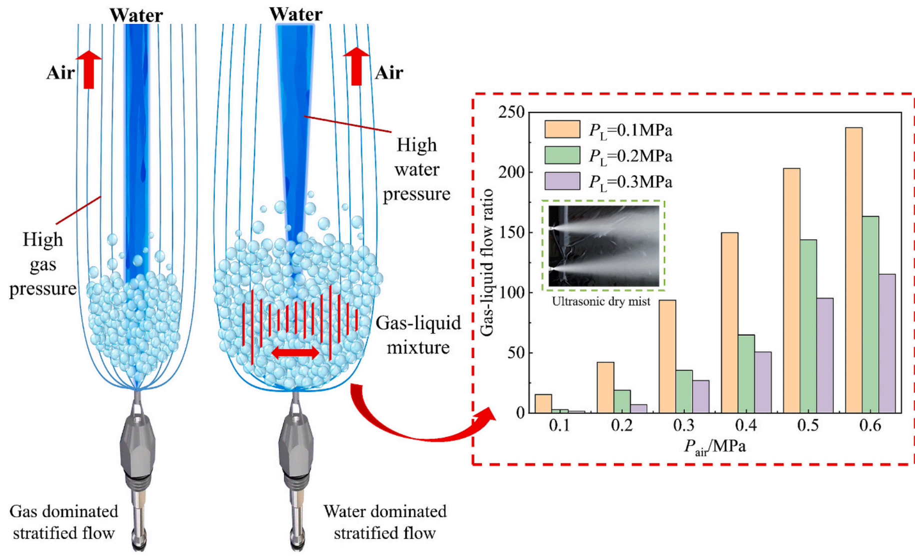

Fig. 3. Nozzle gas-to-liquid flow ratio.

Table 1
Test and fitting formula calculation results of nozzle flow.

Working condition number	P_{air}/MPa	P_L/MPa	Q_1/(L·min^{-1})		Deviation/%	Q_2/(L·min^{-1})		Deviation/%
			Fitting	Experiment		Fitting	Experiment	
1	0.2	0.2	40.44	48.33	16.32	2.31	2.50	7.51
2	0.2	0.3	50.65	55.83	9.28	2.61	2.67	2.21
3	0.3	0.2	72.07	71.67	0.55	1.67	2.02	17.24
4	0.3	0.3	82.27	77.50	6.16	2.03	2.20	7.78
5	0.4	0.2	92.19	98.33	6.25	1.26	1.50	15.69
6	0.4	0.3	102.39	99.17	3.25	1.62	1.70	4.74

the air pressure reaches approximately 0.5 MPa, this downward trend tends to level off, indicating an inflection point.

The principle of fluid ultrasonic nozzle atomization is that high-speed flowing gas collides with the resonance cavity. Under the impact of the gas, the resonance cavity generates ultrasonic waves, which act on the droplets to atomize the droplets into smaller droplets. According to the experimental data of nozzle gas-liquid flow rate, the nozzle gas-to-liquid flow ratio changes with the inlet air pressure as shown in Fig. 3.

The gas-liquid flow ratio reaches its maximum value when the inlet water pressure is 0.1 MPa and the inlet air pressure is 0.6 MPa. A regression analysis was performed on the experimental data, and the relationship between gas-liquid flow rate and gas-liquid pressure were obtained as shown in Eqs. (1) and (2), respectively.

$$Q_1 = 149.37 + 102.08P_L - \frac{36}{P_{air}} + \frac{2.07}{P_{air}{}^2} \tag{1}$$

$$Q_2 = 1.10 + 0.89\ln(P_L) + \frac{0.74}{P_{air}} - \frac{0.04}{P_{air}{}^2} \tag{2}$$

To assess the accuracy of the aforementioned nozzle flow formula, the test also measured the nozzle flow under other working conditions and compared it with the calculation results of the fitting formula, as shown in Table 1. The R^2 of Eq. (1) is 0.985 and the R^2 of Eq. (2) is 0.988, calculated by regression analysis. Furthermore, Eq. (1) is valid when the air and water pressures are between 0.1 and 0.6 MPa. As can be seen from Table 1, the relative error between the nozzle flow experimental measurement results and the calculation results of the fitting formula is less than 18%, and the error of some results is even less than 5%. The above formula can be used for the theoretical calculation of the ultrasonic dry fog nozzle flow.

2.2. Nozzle atomization particle size and droplet movement speed

The nozzle droplet size is a primary determinant of spray performance. It is directly related to the distribution of the liquid, the coverage effect and the uniformity of the spray. The size spectrum of droplets aligns with dust particle dimensions and is more likely to collide and combine. According to the literature, the calculation formula for the average droplet size Dc of the internal mixing air atomization spray is as follows [35,36]:

$$D_C = 585.21\left(\frac{1}{V_r}\sqrt{\frac{\sigma}{\rho_L}}\right) + 5.97\left(\frac{\mu_L}{\sqrt{\sigma\rho_L}}\right)^{0.45}\left(1000\frac{Q_L}{Q_g}\right)^{1.5} \tag{3}$$

Where D_C is the average droplet diameter, m; V_r is the gas-to-liquid slip velocity within the mixing chamber, m/s; σ is the liquid surface tension coefficient, 10^{-5} N/cm ; μ is the viscosity coefficient of the liquid, Pa·s ; ρ_L is the density of the liquid, g/cm^3 ; Q_L/Q_{air} is the liquid-gas flow ratio. Q_L and Q_{air} are the volume flow rates of the liquid and air, m^3/s. Select the relevant parameters of water under standard conditions, $\sigma = 72 \times 10^{-5}$ N/cm, $\mu = 0.00982$ Pa·s and $\rho_L = 1.0$ g/cm^3 substitute them into formula (3) to simplify and obtain formula (4):

$$D_C = \frac{4965.7}{V_r} + 0.284\left(1000\frac{Q_1}{Q_g}\right)^{1.5} \tag{4}$$

The relative flow rate of gas and liquid phases can be calculated using the following formula:

$$V_r = \sqrt{\left(V_g \sin\alpha\right)^2 + \left(V_g \cos\alpha - V_l\right)^2} \tag{5}$$

Where, V_{air} and V_L are the air and water flow rates, m/s; α is the angle between the air and water flow rates, °. Substituting the relationship between flow rate, flow rate and cross-sectional area $V = Q/A$.

Substitute Eq. (4) into Eq. (5) to obtain Eq. (6).

$$D_c = \frac{4965.7}{\sqrt{\left(\frac{Q_{air}}{A_{air}}\sin\alpha\right)^2 + \left(\frac{Q_{air}}{A_{air}}\cos\alpha - \frac{Q_L}{A_L}\right)^2}} + 0.284\left(1000\frac{Q_L}{Q_{air}}\right)^{1.5} \quad (6)$$

Where, A_L is the liquid orifice flow area, m^2 ; A_{air} is the gas orifice flow area, m^2. The diameter of the gas orifice flow area of the nozzle is 0.5 mm, so $A_{air} = 7.854 \times 10^{-7}$ m^2. The diameter of the liquid orifice flow area is 1.0 mm, $A_L = 3.142 \times 10^{-6}$. The angle between the gas orifice flow area and the liquid orifice flow area is $\alpha = 90^\circ$. Substituting the relevant data into formula (6).

$$D_c = \frac{4965.7}{\sqrt{\left(1.273 \times 10^6 Q_{air}\right)^2 + \left(3.182 \times 10^5 Q_L\right)^2}} + 0.284\left(1000\frac{Q_L}{Q_{air}}\right)^{1.5} \quad (7)$$

In this study, the droplet velocity in the stable flow area is selected to represent the average velocity of the droplets in the action area. The Doppler phase interferometer (PDI) is used for measurement, and the 1stOpt software is used to perform regression analysis on the test data. The relationship between the droplet velocity $U_d$of the ultrasonic dry fog nozzle and the inlet liquid pressure and the inlet gas pressure is shown in formula (8):

$$U_d = 5.83 - 0.16P_L + 14P_{air} \quad (8)$$

3. Mathematical model for spray dust capture efficiency

3.1. Model assumptions

The airflow purification curtain set up in the underground coal mine tunnel is taken as an example for analysis. The cross-section of a large coal mine tunnel is generally more than 10 m [37,38]. A row of ultrasonic atomization nozzles with an interval of about 0.5 m is arranged at the top of the tunnel. The spray is sprayed along the wind flow at a 90° perpendicular angle to the wind flow. The joint action of multiple nozzles can make the mist flow evenly cover the entire tunnel cross-section. Under the joint action of wind flow and gravity, the droplets spray from the top of the tunnel to the ground of the tunnel. The high-speed moving water mist covers the entire tunnel cross-section. The droplets reduce the dust concentration in the tunnel by colliding, capturing and settling coal dust. The following assumptions are adopted. They simplify the methodology and enable model development.

(1) The droplet size is the same and is evenly distributed in the mist flow action area;
(2) Each droplet captures dust particles independently and does not affect each other;
(3) The dust exhibits slip-free suspension with uniform tunnel cross-sectional distribution.

3.2. Development of a mathematical model for dust capture efficiency

The wind speed in the tunnel of coal mine with wind purification water curtain is usually low. Within the scope of the air flow purification water curtain, the speed of droplet movement is much greater than that of the air flow and dust. Therefore, it can be approximately considered that the relative speed between the mist droplets and the wind flow and dust is the speed of the mist droplets. Formula (9) can be obtained.

$$U_{dg} = U_0 = U_d \quad (9)$$

In the formula, μ_g represents the gas viscosity coefficient, Pa·s. U_0 is the relative velocity of the dust particle and the droplet in the undisturbed upstream gas, m/s. U_{dg} represents the relative speed between the droplet and the airflow, m/s. Q_L represents the overall volume content of the droplet, m^3/s. When evaluating the effectiveness of dust suppression measures, dust reduction efficiency serves as a crucial quantitative indicator. Huitian et al. proposed a widely applied formula for calculating dust reduction efficiency in dust control using air-water mixing nozzles, which is expressed as follows [39]:

$$\eta = 1 - \exp\left[\frac{-3Q_L x}{2U_g A D_C \times 10^{-6}} B_0 \times \left(\frac{B D_p^2 \rho_p U_d \times 10^{-6}}{B D_p^2 \rho_p U_d \times 10^{-6} + 6.3\mu_g D_C}\right)^2\right] \quad (10)$$

In the dust removal mechanism (mainly including inertial collision, interception and diffusion), the particle size of pulverized coal greater than 1 μm is mainly inertial collision dust removal. In order to simplify the study, when analyzing the spray dust reduction mechanism, only the inertial collision effect between dust and droplets is considered, then $B_0 = B = 1$ in formula (10). Take the real density of pulverized coal on site $\rho_p = 1.05 \times 10^3$ kg/m^3, take the dynamic viscosity coefficient of air $\mu_g = 1.8 \times 10^{-5}$ Pa·s, substitute the above parameters into formula (10), it can be simplified to formula (11):

$$\eta = 1 - \exp\left[\frac{-3Q_L x}{2U_g A D_C \times 10^{-6}}\left(\frac{1}{1 + \frac{0.108 D_C}{D_p^2 U_d}}\right)^2\right] \quad (11)$$

3.3. Analysis of factors affecting dust capture efficiency

In order to verify the effectiveness of the mathematical model for dust capture efficiency and compare the error between the theoretical dust capture efficiency and the actual dust capture efficiency of the mathematical model [40,41], an ultrasonic dry fog purification device was installed in a state-owned coal mine in Shanxi, China for on-site testing. The droplet diameter D_c and U_d in formula (10) can be replaced by formula (3) and formula (8) respectively, and the two can be expressed as expressions of air flow and liquid flow. Since the substitution will make formula (11) too long, the original form of formula (11) is retained. The width of the underground coal mine tunnel used for analysis and calculation is 4 m. Since there is a press-in air duct in the coal mine excavation tunnel, 6 ultrasonic atomization nozzles are evenly arranged in the middle of the tunnel top, and the interval between the nozzles is 0.5 m. Then in formula (6), the Q_L is equal to $6Q_1$. In the field application, the ultrasonic dry fog device is at an angle of 90° with the wind direction of the tunnel, and the effective spray distance is short. The effective action area length of the ultrasonic atomization nozzle measured on site is x ≈ 4 m. Because the ultrasonic dry fog nozzle is equipped with a resonance cavity, it can break the droplets into tiny droplets through ultrasonic action, but in actual engineering applications, the hydrophobicity of coal dust is different, which affects the dust capture efficiency. Since coking coal has poor hydrophobicity, it is difficult for droplets to capture dust. Therefore, the ultrasonic dry fog nozzle is introduced to correct the dust reduction of coking coal. K is 0.96. Substituting the above data into formula (11) to simplify it, we get formula (12).

$$\eta = 1 - \exp\left[\frac{-2KQ_1}{D_C \times 10^{-6}}\left(\frac{1}{1 + \frac{0.108 D_C}{D_p^2 U_d}}\right)^2\right] \quad (12)$$

In formula (12), D_P refers to a specific dust particle size, and the spray dust capture efficiency pertains to dust of this particle size, which is known as classification efficiency. For a particular nozzle, the main factors influencing the classification efficiency of spray dust reduction are the water supply flow rate Q_L, the air supply flow rate Q_{air}, and the dust particle size D_P. According to formula (12), the relationship curve between each influencing factor and the classification efficiency is

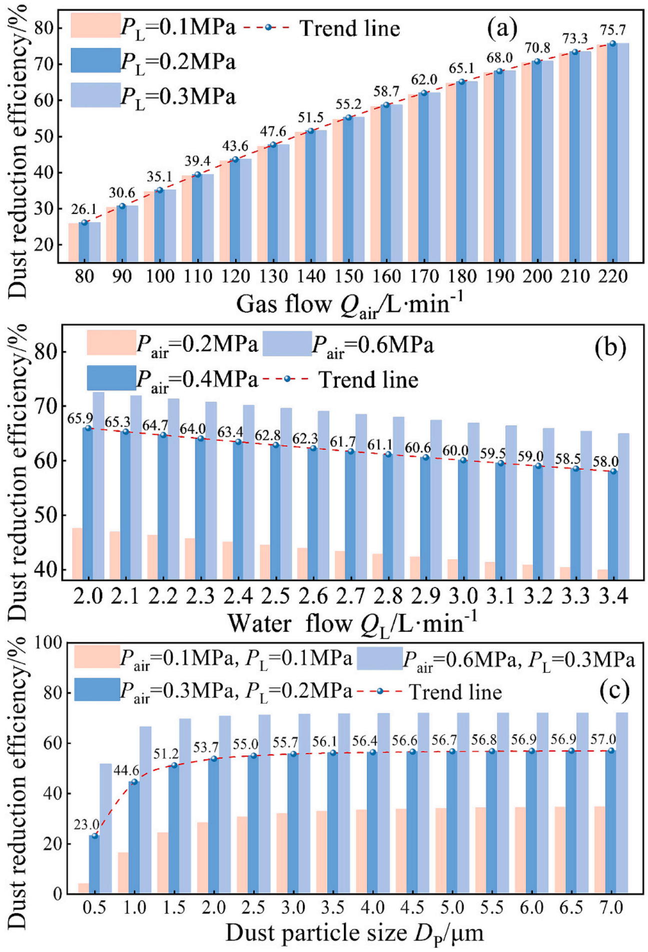

Fig. 4. Relationship curve between classification efficiency and air supply flow, water supply flow, and dust path.

drawn using Origin software, as shown in Fig. 4.

The relationship curve between classification efficiency and inlet air flow rate is shown in Fig. 4(a). The dust particle size is set to 5 μm during analysis. It can be seen that when the inlet water flow rate is fixed, the classification efficiency increases with the increase of inlet air flow rate. From the relationship between droplet size and inlet air flow rate, it can be seen that when the inlet water flow rate remains unchanged, as the inlet air flow rate increases, the droplet size continues to decrease and the droplet concentration continues to increase, thereby improving the inertial collision capture efficiency of the droplets. It can also be seen from the figure that when the inlet air flow rate is fixed, the classification efficiency decreases as the water inlet flow rate increases.

Keeping the dust particle size $D_P = 5$ μm unchanged, Fig. 4(b) shows the relationship between classification efficiency and water flow rate. When the inlet air flow rate remains unchanged, as the inlet water flow rate increases, the characteristic of classification efficiency is an initial decrease and subsequent stabilization. For ultrasonic dry fog spray, when the inlet air flow rate is fixed, when the inlet water flow rate continues to increase, on the one hand, the water content per unit space is increased, but on the other hand, the droplet size also increases.

In Fig. 4 (c), with the inlet water flow fixed at 2.0 L/min, the relation between the classification efficiency and the size of dust particles is analyzed. It is shown that when the air flow remains constant, with the growth of dust particle size, the classification efficiency rises, though the increase slows down once the particle size exceeds a certain value. Particularly, when the inlet air volume reaches 6.0 m^3/h, the classification efficiency of dust particles with particle size greater than 1.0 μm exceeds 50%; When the particle size is less than 1.0 μm, the efficiency

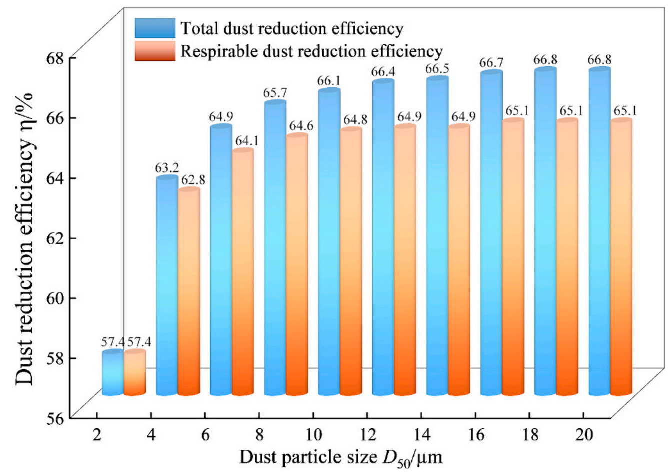

Fig. 5. Relationship between dust reduction efficiency and dust median particle size D_{50}.

decreases sharply. The dust particle size of coal mining working surface generally obeys the Rosin-Rammler distribution, which is expressed as:

$$R(D_P) = \exp[-\ln 2(D_P/D_{50})^s] \quad (13)$$

Where, $R(D_P)$ denotes the sieve retention rate; D_{50} represents the mass median diameter of the particulate matter, μm ; s is uniformity index for particle size distribution. Higher s values indicate a narrower particle size distribution span, while lower s values signify a broader distribution span. The s value of the dust on the coal mining working surface is generally between 0.8 and 1.8. By taking the derivative of formula (13), the frequency distribution of the dust can be obtained as follows:

$$f(D_p) = s\ln 2 D_{50}^{-s} D_p^{s-1} \exp[-\ln 2(D_p/D_{50})^s] \quad (14)$$

Where, $f(D_P)$ is the frequency of dust with different particle sizes. The expression in formula (12) only obtains the classification efficiency of spray dust reduction. For the evaluation of the ultrasonic dry fog spray dust reduction effect, the overall dust capture efficiency and the respirable dust removal efficiency are generally used. According to formula (13) and formula (14), the expression of the overall dust capture efficiency η' can be obtained as follows:

$$\eta' = \int_0^{+\infty} \eta f(D_p) \mathrm{d}D_p \quad (15)$$

Similarly, the dust capture efficiency η'' of respirable dust can be expressed as:

$$\eta'' = \int_0^{7.07} \frac{\eta f(D_p)}{1 - R(7.07)} \mathrm{d}D_p \quad (16)$$

Combining Eqs. (13) to (16), we can derive the relationship between the overall dust capture efficiency the respirable dust removal efficiency, the operating parameters (air flow and water flow) and the dust particle size parameters (median particle size and distribution index). After analysis, it was found that the relationship between overall dust removal efficiency and the respirable dust removal efficiency and each operating parameter showed a similar change pattern to the classification efficiency, which will not be repeated here.

Under the conditions of air supply volume of 180 L/min, water supply volume of 2.45 L/min and dust particle size distribution index s fixed at 1.5, the relationship curve between dust capture efficiency of overall dust and respirable dust and median particle size D_{50} shown in Fig. 5 is obtained by solving Eqs. (15) and (16) with MATLAB. It is indicated that with the above - mentioned operating conditions and s value remaining unchanged, both kinds of dust collection efficiencies

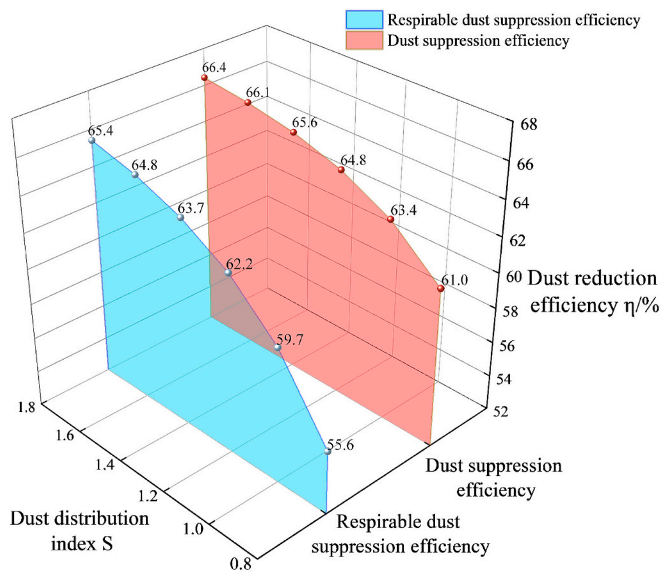

Fig. 6. Relationship between dust capture efficiency and dust distribution index s.

rise as D_{50} goes up. Also, the overall dust collection efficiency is constantly higher than that of the respirable dust, and the disparity between them enlarges as D_{50} increases. It can be attributed to the fact that the classification effectiveness rises as the dust particle size grows: when the *s* value remains unchanged, an increase in D_{50} will cause the average dust particle size to increase, thereby enhancing the two types of efficiencies. More importantly, the increase of D_{50} will reduce the mass proportion of respirable dust in overall dust, which will directly lead to the widening gap between the overall dust capture efficiency and the respirable dust removal efficiency.

When the air flow rate, water flow rate and median particle size of dust (10 μm) are fixed, Fig. 6 shows that the overall dust removal efficiency and respiratory dust removal efficiency increase with the particle size distribution index rises, and the gap between the two continues to narrow. This is because the increase of distribution index leads to the narrower distribution range of dust particle size, and the particles are more concentrated near the median particle size (10 μm), which reduces the proportion of fine particles ($dp < 2$ μm) in overall dust and respirable dust, thus improving the two kinds of dust capture efficiency. Meanwhile, the concentration of particle size distribution and close to the10μm median particle size, which leads to the reduction of the difference between the overall dust capture efficiency and the respirable dust capture efficiency.

4. Field application

4.1. Design of dry fog purification device

The overall design rationality of purification equipment directly affects its performance and manufacturability. The core of the design is to reasonably arrange the installation positions of the gas-liquid pipelines and nozzles so that they can operate normally and are easy to install. The dry fog purification equipment is mainly composed of a rod body and sealing screws, as shown in the Fig. 7. Among them, the air supply pipeline and water supply pipeline that are not connected to each other are arranged inside the rod body, the air inlet hole connecting the air supply pipeline and the water inlet hole connecting the water supply pipeline are arranged in the middle of the rod body, and the installation holes for installing the nozzles are evenly arranged in the rod body. Inside the rod body, the nozzle installation hole is connected to both the air supply pipeline and the water supply pipeline. The sealing screws are installed at both ends of the rod body to form a closed space inside the rod body, and the nozzle outlet is the only air and water discharge port, as shown in Fig. 7.

4.2. Equipment installation and experimental plan

The coal mine machinery heading face is a primary dust source in coal mines. The fully-mechanized excavation face is a semi-enclosed limited space with large and concentrated dust production [42,43]. Its main dust source is the cutting dust produced by the fully-mechanized excavator at the end of the excavation.

At the same time, the dust dispersion of the coal mine machinery heading face is high, and fine respirable dust is difficult to capture and settle, which is extremely harmful to the human body. Through the investigation of the actual engineering site and comprehensive

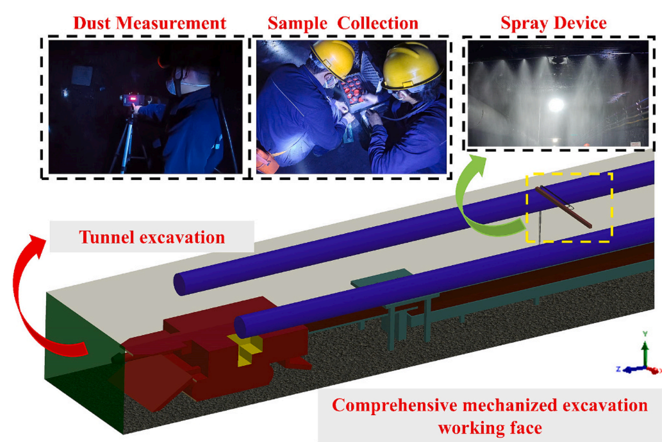

Fig. 8. On-site application of wind flow dry fog device.

Fig. 7. Schematic diagram of dry fog device.

Table 2
Dust capture efficiency at each measuring point.

Dust reduction device	Sampling times	Theoretical overall dust capture efficiency (%)	Actual overall dust capture efficiency (%)	Theoretical respirable dust capture efficiency (%)	Actual respirable dust capture efficiency (%)
Ultrasonic dry fog dust capture device	1st	67	61.56	65	55.97
	2nd	67	63.78	65	56.21
	3rd	67	61.31	65	54.37
	4th	67	59.05	65	57.64

consideration of the experimental sampling conditions, the inlet air pressure is set to 0.5 MPa and the inlet water pressure is set to 0.1 MPa. The nozzle spacing is 50 cm, and the spray can cover the entire tunnel. The control box starts and stops the ultrasonic dry fog dust reduction device. Use FCC-25 explosion-proof dust sampler to collect dust samples on site according to the actual situation on site. The sampling point was 5 m away from the purification device, located in the return air flow direction of the tunnel, and about 300 m away from the comprehensive excavation working face. Two sampling points were set before and after the dry fog device, and the samples were collected three times to measure the dust concentration distribution. The dust concentration changes before and after dust reduction were compared. The on-site application diagram is shown in Fig. 8.

4.3. Field measurement results

According to the installation plan of the device, the dust sampling points were arranged according to the test plan, and the dust capture performance test experiment of the device was carried out. After the excavator was working, the device reached the required pressure, and the influence of other irrelevant factors on the test results was eliminated. The dust samplers at each measuring point were turned on for 2 min, then measure the dust concentration before and after the purification device. The filter membrane that collected the dust was dried and weighed to obtain the dust concentration at each measuring point [44,45]. To assess the device's respirable dust capture efficiency, we used an LS-13320 laser diffraction analyzer to measure the particle size distribution of particulate matter. The results of overall dust capture efficiency and dust removal efficiency of respirable dust at different measuring points of the device are shown in Table 2.

The airflow dry fog dust reduction device significantly improved tunnel dust concentration. The theoretical overall and respirable dust reduction efficiencies were 67% and 65%, respectively. The actual average overall dust capture efficiency was 61.42%, and the average respirable dust capture efficiency was 56.05%. Actual efficiencies were slightly lower than theoretical values. The discrepancy is attributed to wind-induced drift of fine droplets, preventing full dust contact. Airflow fluctuations and droplet drift at the underground coal mine working face are also key factors leading to the difference between theoretical and measured dust removal efficiency. However, the error is less than 20%, indicating the model provides a valid theoretical reference for field application.

5. Conclusions

(1) The atomization characteristics and dust suppression efficiency of ultrasonic dry fog nozzles are significantly affected by the coupling of gas and liquid pressure. Increased gas supply pressure leads to increased gas flow rate, decreased water flow rate, finer droplet size, and increased range; increased water supply pressure causes a simultaneous increase in both gas and liquid flow rates, but has a weaker impact on the atomization angle and range. Droplet size continuously decreases with increasing gas-liquid ratio.

(2) Based on experimental results, a predictive model and graded efficiency for the dust suppression efficiency of ultrasonic dry fog nozzles were established. The graded efficiency model systematically revealed the influence of air supply, water supply, and dust particle size on the collection efficiency. This model can provide a theoretical reference for the design and operation parameter adjustment of ultrasonic dry fog spraying systems under different mining conditions.

(3) Field application showed theoretical overall and respirable dust reduction efficiencies of 67% and 65%. Actual average efficiencies were 61.42% for overall dust and 56.05% for respirable dust. The actual values are slightly lower than theoretical predictions, but the error is below 20%, demonstrating that the mathematical model provides a reliable theoretical reference for the on-site application of ultrasonic dry fog nozzles.

CRediT authorship contribution statement

Xinzhe Wang: Writing – original draft, Validation, Methodology. **Pengfei Wang:** Writing – review & editing, Conceptualization. **Yun Peng:** Funding acquisition. **Yongjun Li:** Data curation. **YaFei Luo:** Visualization. **Shilin Li:** Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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